

# Cross-Taxonomy Label Alignment in Transportation Networks with Prompt-Driven Large Language Models

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**Abstract**—Integrating transportation datasets from diverse sources is critical for comprehensive analysis and intelligent infrastructure planning. A major obstacle to effective integration lies in taxonomy heterogeneity, where datasets differ in points of interest (POI) categories, road type classifications, and land use taxonomies. Addressing these inconsistencies requires cross-taxonomy label alignment, a challenge that remains inadequately explored due to the reliance on costly human annotations and the limited effectiveness of rule-based matching approaches. This paper presents a novel framework that combines prompt-driven pseudo-labeling and multi-view representation learning to address cross-taxonomy label alignment. Large language model (LLM) prompts generate initial semantic pseudo-labels to guide pretraining, while multi-view input representations are constructed by combining DeepWalk-based structural embeddings with encoded source taxonomy labels. The framework is first trained in a semi-supervised manner using pseudo-labels, followed by fine-tuning with a limited amount of human-verified labels to further refine alignment quality. Extensive experiments demonstrate that the proposed method substantially outperforms rule-based and prompt-only baselines, maintaining robust performance even when only 1% of labeled samples are available. These results demonstrate that leveraging LLM-guided semantic supervision alongside structural representations enables robust and scalable cross-taxonomy adaptation, even under severely limited supervision.

**Index Terms**—Cross-Taxonomy Label Alignment, Transportation Data Integration, Prompt-Based Learning, Multi-View Representation Learning, Semi-Supervised Learning, Low-Resource Supervision, Structural Embedding, Large Language Models

## I. INTRODUCTION

Transportation systems generate vast and diverse datasets, ranging from road network infrastructure to facility and land use annotations. Integrating such multi-source data is crucial for applications such as traffic forecasting, infrastructure planning, and intelligent transportation systems [3], [12].

A major obstacle to effective integration lies in taxonomy heterogeneity, where datasets differ in categorizing transportation entities such as points of interest (POIs), road types, and

land use classes. Different sources, such as OpenStreetMap (OSM) and the Federal Highway Administration (FHWA) databases, often define distinct classification standards, leading to inconsistencies that hinder unified data modeling across systems [2].

Existing efforts for cross-dataset integration mainly rely on manual mapping tables or rule-based matching techniques [14], which are labor-intensive and lack scalability to large or dynamically evolving datasets. Some machine learning approaches attempt to align categories based on semantic similarities [6], but they often assume the availability of extensive labeled data or rely on fully supervised learning paradigms. Maintaining robust alignment performance remains a significant challenge in low-resource settings, where only limited human-verified samples are available.

Despite these developments, few studies have systematically addressed the problem of cross-taxonomy label alignment in transportation domains. Most existing work focuses on attribute-level matching, aligning fields such as names or IDs, while overlooking deeper semantic misalignments between classification systems and the complementary structural information embedded in transportation networks [4].

Addressing these inconsistencies requires solving a cross-taxonomy label alignment problem, where labels from a source taxonomy must be mapped to semantically corresponding labels in a target taxonomy. Figure 1 illustrates the overall problem setting. Source taxonomy labels (e.g., OSM road categories) and target taxonomy labels (e.g., FHWA road classifications) may differ significantly in granularity and semantics, and the correspondence between them is not necessarily one-to-one.

To tackle this challenge, we propose a novel framework that leverages large language models (LLMs) to generate semantic pseudo-labels, integrates structural representations derived from network topology, and refines the alignment with limited human-labeled data. By combining semantic and structural views, the proposed framework achieves robust cross-taxonomy adaptation even under severely limited supervision.

The main contributions of this work are summarized as

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. 2021R1C1C1006405). (Corresponding author: Inhi Kim).

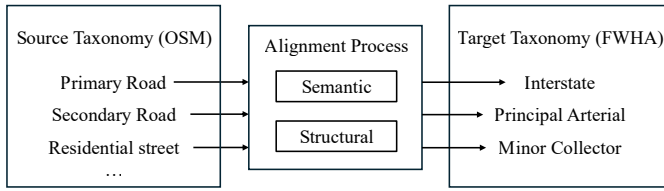


Fig. 1. Illustration of the cross-taxonomy label alignment task. Source taxonomy labels are mapped to target taxonomy labels through an alignment process guided by semantic supervision (prompt-derived pseudo-labels) and structural context (network embeddings).

follows:

- We systematically formulate the problem of cross-taxonomy label alignment in heterogeneous transportation datasets and highlight the limitations of existing integration methods.
- We propose a novel LLM-guided framework that combines prompt-driven pseudo-labeling, structural feature learning, and limited-supervision fine-tuning to achieve robust alignment.
- We conduct extensive experiments demonstrating that the proposed method substantially improves alignment performance over prompt-only baselines and direct supervised training, particularly under low-resource conditions, establishing its practicality for scalable transportation data integration.

## II. RELATED WORK

### A. Cross-Taxonomy Label Alignment in Transportation

Integrating transportation datasets from diverse sources, such as OpenStreetMap (OSM) and governmental road inventories, is crucial for comprehensive transportation analysis. However, taxonomy heterogeneity introduces inconsistencies that hinder cross-dataset modeling. Existing efforts often rely on manual mapping tables or heuristic matching rules [1], which are labor-intensive and lack scalability. While supervised ontology alignment techniques have been extensively studied in broader knowledge domains [8], systematic solutions for cross-taxonomy label alignment in transportation settings remain scarce, particularly under limited supervision.

### B. Prompt-Based Pseudo-Labeling and Weak Supervision

Recent advances in large language models (LLMs) have enabled prompt-based pseudo-labeling for various machine learning tasks, including zero-shot classification and weak supervision [5], [9]. These methods leverage natural language prompts to generate initial guidance, reducing dependence on manual labeling. While prompt learning has been successfully applied to structured data tasks [10], [11], its potential for cross-taxonomy label alignment, especially within hierarchical and domain-specific transportation datasets, remains underexplored.

### C. Structural Representation Learning in Transportation Networks

Structural feature learning approaches, such as node embeddings generated by DeepWalk [7] and graph neural networks (GNNs) [13], [15], have been widely applied to transportation network modeling. These methods capture topological relationships between road segments, facilities, and intersections, enabling improved prediction and classification tasks. Incorporating structural embeddings has proven effective in traffic flow prediction and road type classification. However, their integration into cross-taxonomy label alignment tasks, particularly in conjunction with semantic pseudo-labeling, has not been systematically investigated.

### D. Positioning of This Work

Building upon these prior efforts, this paper proposes a novel framework that integrates prompt-driven semantic pseudo-labeling and structural feature learning to address the cross-taxonomy label alignment problem in transportation datasets. Unlike existing approaches that either rely heavily on manual mappings, assume one-to-one label correspondences, or require extensive labeled data, the proposed method fuses semantic and structural views under a semi-supervised pre-training and fine-tuning paradigm. This enables robust and scalable alignment even under severely limited human supervision, bridging a critical gap in transportation data integration research.

## III. METHODOLOGY

This section introduces our label alignment approach that combines semantic supervision from large language models (LLMs) and structural features derived from road network embeddings. The proposed method is designed to support taxonomy-level label alignment in transportation datasets under limited human annotation.

### A. Framework Overview

Figure 2 illustrates the overall architecture of the proposed framework. Semantic features are derived from label embeddings and LLM-guided pseudo-labels generated from source taxonomy labels. Structural features are obtained using DeepWalk from the road network. These two views are concatenated to form multi-view inputs for training. During the first stage, the model is pretrained on pseudo-labeled samples, with semantic masking applied to encourage reliance on structural context. In the second stage, fine-tuning is performed using a small amount of human-labeled data.

### B. Prompt-Based Pseudo-Label Generation

To address the lack of human-labeled alignment pairs, we generate pseudo-labels at the taxonomy level using LLMs (e.g., ChatGPT). The overall process for prompt-driven pseudo-label generation is illustrated in Figure 3. Multiple prompts are issued to the LLM for each source taxonomy label to retrieve candidate mappings, which are then aggregated via majority voting to determine the final pseudo-label. For

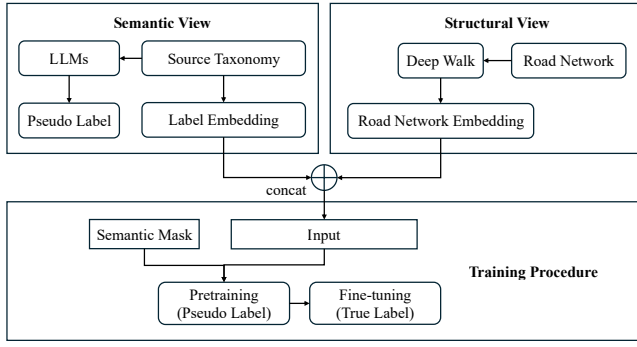


Fig. 2. Overview of the proposed label alignment framework. Semantic features are obtained from source labels and LLM-generated pseudo-labels; structural features are derived from DeepWalk on the road graph. A two-stage training procedure (pretraining + fine-tuning) is applied. Semantic masking is used to improve robustness.

each source taxonomy label  $l_A \in \mathcal{L}_A$ , we define a set of  $k$  prompt variants  $\mathcal{P} = \{p_1, p_2, \dots, p_k\}$  designed to elicit its most semantically aligned counterpart in the target taxonomy.

Each prompt  $p_i$  is used to query the LLM, returning a candidate label mapping:

$$\phi_i(l_A) = \text{LLM}(p_i, l_A) \quad (1)$$

A majority voting mechanism is applied to the  $k$  outputs:

$$\phi(l_A) = \text{mode}(\{\phi_i(l_A)\}_{i=1}^k) \quad (2)$$

This taxonomy-level mapping  $\phi : \mathcal{L}_A \rightarrow \mathcal{L}_B$  is then used to assign pseudo-labels to all nodes in the dataset annotated with  $l_A$ . In cases without consensus (e.g., tie in votes), the corresponding samples are excluded from the pre-training stage.

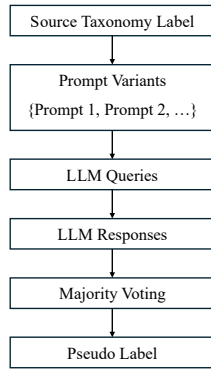


Fig. 3. Pipeline for prompt-driven pseudo-label generation. Multiple prompt variants are issued per source label, and responses from the LLM are aggregated via majority voting to determine the final pseudo-label assignment.

### C. Structural Embedding Generation

To encode topological information, we generate structural node embeddings using DeepWalk. The algorithm simulates  $w$  random walks of length  $l$  for each node  $v \in V$ , producing node sequences over which a Skip-gram model is trained. The

objective is to maximize the likelihood of context nodes given a center node:

$$\max \sum_{i=1}^{|V|} \sum_{-c \leq j \leq c, j \neq 0} \log p(v_{i+j} | v_i) \quad (3)$$

This results in an embedding vector  $\mathbf{h}_v^{struct} \in \mathbb{R}^d$  for each node  $v$ , capturing structural relationships.

### D. Semantic Masking for Robust Learning

We apply semantic masking during pretraining to reduce overfitting to deterministic pseudo-label mappings and promote structural learning. In each training batch, a random subset  $M \subset V$  of nodes has their semantic inputs removed:

$$\mathbf{x}_v = \begin{cases} [\mathbf{0}; \mathbf{h}_v^{struct}], & \text{if } v \in M \\ [\mathbf{h}_v^{sem}; \mathbf{h}_v^{struct}], & \text{otherwise} \end{cases} \quad (4)$$

This forces the model to rely on structural features and improves generalization under noisy or low-confidence pseudo-labels. Semantic masking also enhances robustness when the label spaces of  $T_A$  and  $T_B$  differ significantly.

### E. Two-Stage Training Strategy

The model is trained using a two-stage procedure, consisting of pretraining on pseudo-labeled samples and fine-tuning on a small set of human-verified labels.

**Stage 1: Pretraining on Pseudo-Labels.** In the first stage, the model is trained on the pseudo-labeled set  $V_{pseudo}$  using cross-entropy loss:

$$\mathcal{L}_{pseudo} = - \sum_{v \in V_{pseudo}} \log p(y_v^{pseudo} | \mathbf{x}_v) \quad (5)$$

where  $\mathbf{x}_v$  is the fused input feature and  $y_v^{pseudo}$  is the pseudo-label assigned via the LLM prompt process. The model is optimized solely on this loss during pretraining.

**Stage 2: Fine-Tuning on Human Labels.** After pretraining converges, the model is fine-tuned on a smaller set of manually annotated nodes  $V_{labeled}$ . A separate cross-entropy loss is used:

$$\mathcal{L}_{fine} = - \sum_{v \in V_{labeled}} \log p(y_v | \mathbf{x}_v) \quad (6)$$

During fine-tuning, the model parameters are updated based on  $\mathcal{L}_{fine}$ , correcting errors introduced by noisy pseudo-labels and adapting the model more closely to human supervision.

**Remark.** The two stages are trained sequentially rather than jointly: first minimizing  $\mathcal{L}_{pseudo}$  until convergence, then minimizing  $\mathcal{L}_{fine}$  starting from the pretrained model parameters. No combined or multi-task loss is used.

## IV. EXPERIMENTAL SETUP

This section describes the datasets, preprocessing procedures, model configuration, evaluation metrics, and baseline methods used to validate the proposed cross-taxonomy label alignment framework.

### A. Datasets

We evaluated our method using transportation network data from two heterogeneous labeling sources:

- **OpenStreetMap (OSM):** An open-source geographic database providing road segment information with diverse ‘highway’ tags, used as the source taxonomy  $T_A$ .
- **Federal Highway Administration (FHWA) Functional Classification:** A standardized classification system for roadways in the United States, serving as the target taxonomy  $T_B$ .

Each road segment is associated with an OSM ‘highway’ tag and an FHWA functional class label, either through existing annotations or manual verification. The goal is to predict the correct  $T_B$  label based solely on  $T_A$ -based semantic and structural cues, without relying on predefined mappings.

### B. Preprocessing

The transportation networks are first constructed as undirected graphs, where each node represents a road segment and edges denote spatial connectivity based on physical adjacency. Structural features are extracted by applying DeepWalk on the constructed graphs, using an embedding dimension of  $d = 128$ , walk length  $l = 40$ , window size  $c = 5$ , and number of walks per node  $w = 10$ . To generate semantic supervision, for each unique source label in the OSM taxonomy  $T_A$ , multiple natural language prompts are designed to elicit corresponding labels in the FHWA taxonomy  $T_B$  via an LLM. Nodes with missing or ambiguous source or target labels are excluded from the dataset to ensure consistency and reliability in evaluation.

### C. Model Implementation and Evaluation

Each node’s input feature is constructed by concatenating its semantic embedding—obtained from one-hot encoded source labels with its structural embedding derived from DeepWalk. A two-layer multilayer perceptron (MLP) is used as the classifier, projecting the fused multi-view inputs into the target taxonomy label space. The model is trained using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 128. Early stopping is applied based on validation loss, with a patience threshold of 20 epochs. Fine-tuning is conducted by initializing the model from the pretrained pseudo-label stage and further training on a small fraction of human-labeled samples (e.g., 10%).

Model performance is evaluated using two metrics: **Accuracy**, which measures the proportion of correctly aligned labels among all evaluated nodes, and **Macro-F1 Score**, which computes the unweighted average of F1 scores across all classes. These metrics assess both overall and class-balanced alignment quality.

### D. Baseline Methods

We compare our proposed framework against four baseline methods to isolate the contributions of prompt-derived semantic supervision, structural feature learning, and multi-view fusion:

TABLE I  
OVERALL PERFORMANCE FOR CROSS-TAXONOMY LABEL ALIGNMENT

| Method                  | Accuracy (%)      | Macro-F1 (%)      |
|-------------------------|-------------------|-------------------|
| <i>Baseline Methods</i> |                   |                   |
| String-Match            | 29.5 ± 0.0        | 20.9 ± 0.0        |
| Prompt-Map              | 54.3 ± 0.0        | 51.8 ± 0.0        |
| <i>Ablation Study</i>   |                   |                   |
| W/o Structure           | 54.8 ± 0.0        | 51.5 ± 0.0        |
| Structure-only          | 68.5 ± 0.5        | 71.7 ± 0.5        |
| <i>Proposed Method</i>  |                   |                   |
| Proposed (Pretrained)   | 46.7 ± 0.4        | 44.8 ± 0.4        |
| Proposed (Fine-tuned)   | <b>79.1 ± 0.4</b> | <b>81.0 ± 0.3</b> |

- **String-Match:** A keyword-based mapping between source and target taxonomy labels without model training.
- **Prompt-Map:** Direct mapping based on LLM-generated pseudo-labels assigned to each source taxonomy label, without further learning or model adjustment.
- **W/o Structure:** A two-layer MLP trained solely on semantic features (source label embeddings) without incorporating any structural embeddings.
- **Structure-Only:** A two-layer MLP trained solely on structural embeddings (DeepWalk), ignoring semantic features.

### E. Prompt-Based Pseudo-Label Generation

To address the scarcity of manually annotated alignment pairs, pseudo-labels are generated at the category level. For each source taxonomy label in  $T_A$ , multiple natural language prompts are issued to an LLM. Responses are aggregated via majority voting to determine a deterministic mapping  $\phi : \mathcal{L}_A \rightarrow \mathcal{L}_B$ . Consequently, all nodes sharing the same source label are assigned the same pseudo-label, ensuring consistency within each category. This strategy highlights the importance of model learning beyond deterministic mappings, particularly when structural cues are incorporated.

## V. RESULTS AND DISCUSSION

This section evaluates the effectiveness of the proposed framework from three perspectives: (1) alignment performance improvement, (2) robustness under limited supervision, and (3) suitability for cross-taxonomy alignment modeling in heterogeneous transportation datasets.

### A. Overall Performance and Structural Contributions

Table I summarizes the cross-taxonomy alignment performance under various training settings, including prompt-only assignment, MLP trained on known taxonomy A labels without structural embeddings, semi-supervised pretraining, and supervised fine-tuning with human-verified data.

Overall, the proposed framework, which combines semi-supervised pretraining with supervised fine-tuning, significantly outperforms the prompt-only baselines across both Accuracy and Macro-F1 metrics. Fine-tuning with human-verified labels leads to substantial performance gains over the

pretraining stage, highlighting the effectiveness of correcting pseudo-label noise through supervised learning.

Analyzing the baselines, String-Match achieves limited performance due to its reliance on literal matching, while Prompt-Map benefits from taxonomy-level semantic similarities, achieving a substantial improvement (54.3% vs. 29.5% in Accuracy).

Beyond overall performance, we further investigate the contribution of structural information. Comparing the "W/o Structure" and "Structure-only" variants reported in Table I, we observe that incorporating DeepWalk-based embeddings improves Accuracy from 54.8% to 68.5% and Macro-F1 from 51.5% to 71.7%. This indicates that network topology provides significant complementary signals beyond category-level semantic information.

These results confirm that while semantic and structural views individually contribute to alignment performance, their integration is crucial for achieving robust cross-taxonomy alignment, as each view captures distinct and complementary aspects of entity relationships.

### B. Robustness under Limited Supervision

Table II and Figure 4 present the model's performance when fine-tuning with varying proportions of human-labeled data.

When 100% of labeled data are available, Direct Supervised and Prompt + Fine-tuning achieve comparable results, with Prompt + Fine-tuning slightly outperforming in Macro-F1 (81.0% vs. 81.1%). As the label ratio decreases, a distinct trend emerges: Direct Supervised performance deteriorates more rapidly, especially in low-resource regimes below 5%.

Notably, at 1% label ratio, Prompt + Fine-tuning achieves 51.3% Accuracy and 50.4% Macro-F1, significantly outperforming Direct Supervised (40.0% Accuracy, 33.4% Macro-F1). This advantage becomes even more pronounced at 0.5% and 0.1% label ratios, where Prompt + Fine-tuning consistently maintains Accuracy above 47%, while Direct Supervised drops below 30%.

These results demonstrate that prompt-enhanced pretraining provides a strong initialization that improves model robustness under limited supervision. Even when fine-tuning with only 1% of labeled samples, the model retains over 90% of its full-data performance, highlighting the framework's practicality for real-world deployment where annotated data are scarce.

Moreover, both String-Match and Prompt-Map baselines remain constant across different label ratios, serving as reference points. The curves in Figure 4 clearly show that Prompt + Fine-tuning consistently surpasses these baselines, especially in the low-resource settings, reinforcing the effectiveness of combining prompt-derived supervision with limited human annotations.

### C. Suitability for Cross-Taxonomy Alignment Modeling

The proposed framework not only improves label alignment accuracy but also demonstrates strong generalization across heterogeneous taxonomy systems without relying on manually designed mappings.

First, the prompt-driven pseudo-labeling process flexibly adapts to varied label vocabularies, requiring no predefined mapping tables between taxonomies. This enables the model to accommodate diverse classification schemes, such as differences between OpenStreetMap (OSM) and FHWA road classes, without extensive manual effort.

Second, the integration of structural embeddings ensures that alignment decisions are informed by spatial and topological relationships between transportation entities. This allows the model to effectively align both major road categories (e.g., Interstate highways) and finer-grained classes (e.g., Minor Arterials and Collectors), as evidenced by the consistent improvements in Macro-F1 scores.

Moreover, the robustness analysis under low-resource conditions demonstrates that the framework retains strong alignment capabilities even with severely limited human supervision, further supporting its scalability to new datasets and evolving taxonomy standards.

Together, these results confirm that the proposed method offers a practical, scalable, and generalizable solution for cross-taxonomy label alignment across heterogeneous transportation datasets.

## VI. CONCLUSION

This paper proposes a prompt-guided and structure-enhanced framework for cross-taxonomy label alignment in heterogeneous transportation datasets. Addressing the challenge of harmonizing diverse classification standards, the method leverages pseudo-labels generated from large language models (LLMs) as semantic supervision, combined with structural embeddings derived from transportation network topology. This design bridges the semantic gap between heterogeneous label spaces without relying on manually curated mappings or extensive human annotation.

Extensive experiments demonstrate that the proposed method substantially outperforms string-matching, prompt-only, and structure-only baselines, and maintains robust alignment performance under low-resource settings. Notably, prompt-guided pretraining provides a strong initialization that enhances model resilience when labeled samples are scarce. The analysis further highlights the critical role of multi-view feature fusion, where semantic and structural representations complement each other to improve alignment accuracy and generalization.

Future work will explore enhancing the framework toward fully unsupervised or dynamically adaptive alignment, incorporating richer external knowledge sources, improving multi-view fusion strategies, and extending the model to accommodate evolving and highly heterogeneous taxonomy standards across multi-source transportation datasets.

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TABLE II  
PERFORMANCE COMPARISON UNDER LOW-RESOURCE SETTINGS: DIRECT SUPERVISED VS. PROMPT-ENHANCED FINE-TUNING

| Label Ratio | Direct Supervised |                | Prompt + Fine-Tuning |                |
|-------------|-------------------|----------------|----------------------|----------------|
|             | Accuracy (%)      | Macro-F1 (%)   | Accuracy (%)         | Macro-F1 (%)   |
| 100%        | 79.1 $\pm$ 0.3    | 81.1 $\pm$ 0.3 | 79.1 $\pm$ 0.4       | 81.0 $\pm$ 0.3 |
| 50%         | 74.9 $\pm$ 0.5    | 77.2 $\pm$ 0.4 | 75.1 $\pm$ 0.2       | 77.3 $\pm$ 0.2 |
| 20%         | 66.7 $\pm$ 0.4    | 68.7 $\pm$ 0.5 | 66.6 $\pm$ 0.3       | 68.7 $\pm$ 0.2 |
| 10%         | 60.0 $\pm$ 0.3    | 61.0 $\pm$ 0.3 | 58.5 $\pm$ 0.4       | 59.7 $\pm$ 0.5 |
| 5%          | 55.5 $\pm$ 0.3    | 56.1 $\pm$ 0.3 | 55.4 $\pm$ 0.1       | 54.6 $\pm$ 0.1 |
| 1%          | 40.0 $\pm$ 0.5    | 33.4 $\pm$ 0.6 | 51.3 $\pm$ 0.1       | 50.4 $\pm$ 0.2 |
| 0.5%        | 35.1 $\pm$ 0.5    | 24.8 $\pm$ 0.8 | 48.2 $\pm$ 0.5       | 47.6 $\pm$ 0.2 |
| 0.1%        | 29.9 $\pm$ 0.7    | 24.3 $\pm$ 0.8 | 47.0 $\pm$ 0.7       | 46.9 $\pm$ 0.4 |

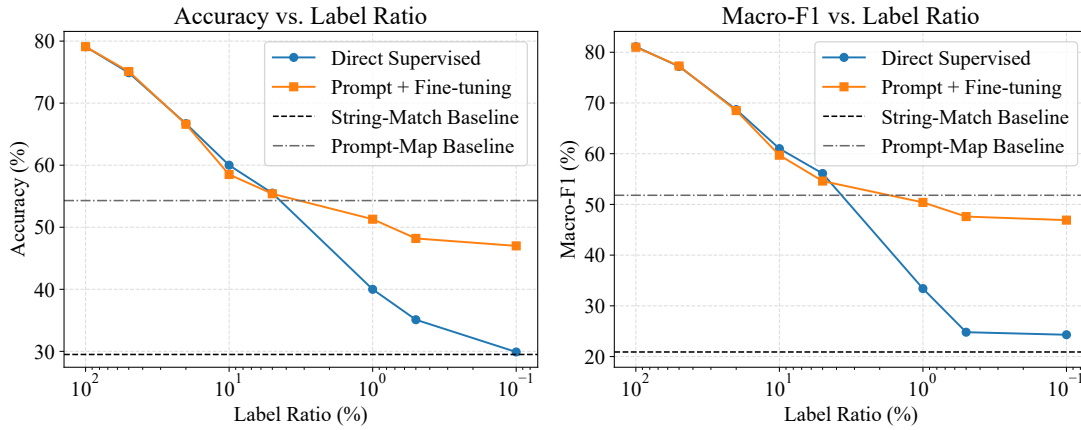


Fig. 4. Performance trends under different fine-tuning label ratios.

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